Mental Health Signal Analysis System:

Evolution from Disorder Detection to Multi-Task NLP Pipeline

<u>A PROJECT REPORT</u>

Submitted By

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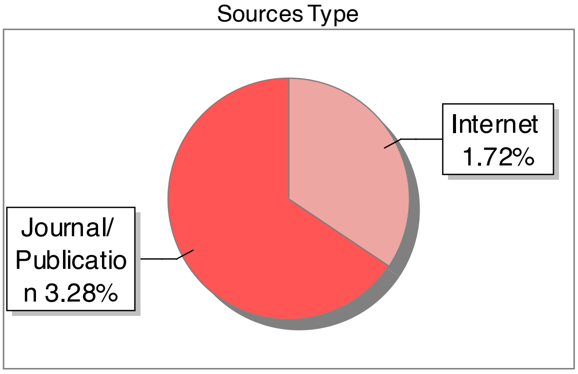

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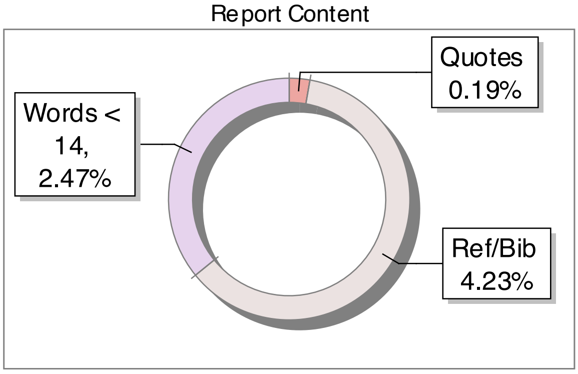

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6.2 Architectural Overview

Mental Wellbeing Guard is composed of three independently deployable components: the Chrome MV3 browser extension, the Hugging Face Space cloud API, and the Dockerized Flask local backend. The extension is the single client; the two backend components are alternative inference targets. The extension's mode manager determines which backend receives inference requests based on the user's selected mode and a health check result in Local Mode.

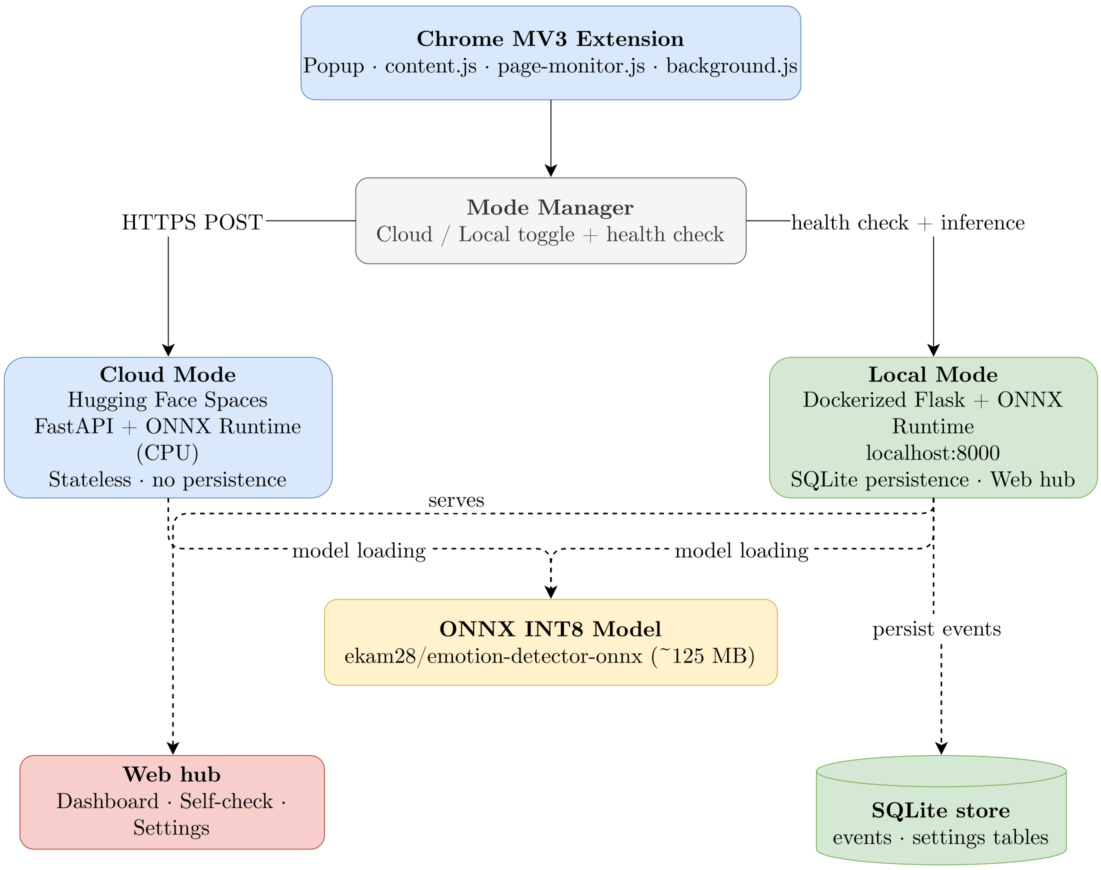

Figure 6.2: Overall Dual-Mode System Architecture

6.3 Cloud and Local Mode Data Flows

In Cloud Mode, comment text extracted by the extension is sent via HTTPS POST to the /batch endpoint of the Hugging Face Space at https://ekam28-emotion-detector-api.hf.space. ONNX Runtime loads the quantised RoBERTa model and returns JSON label probability distributions. No data is persisted in Cloud Mode.

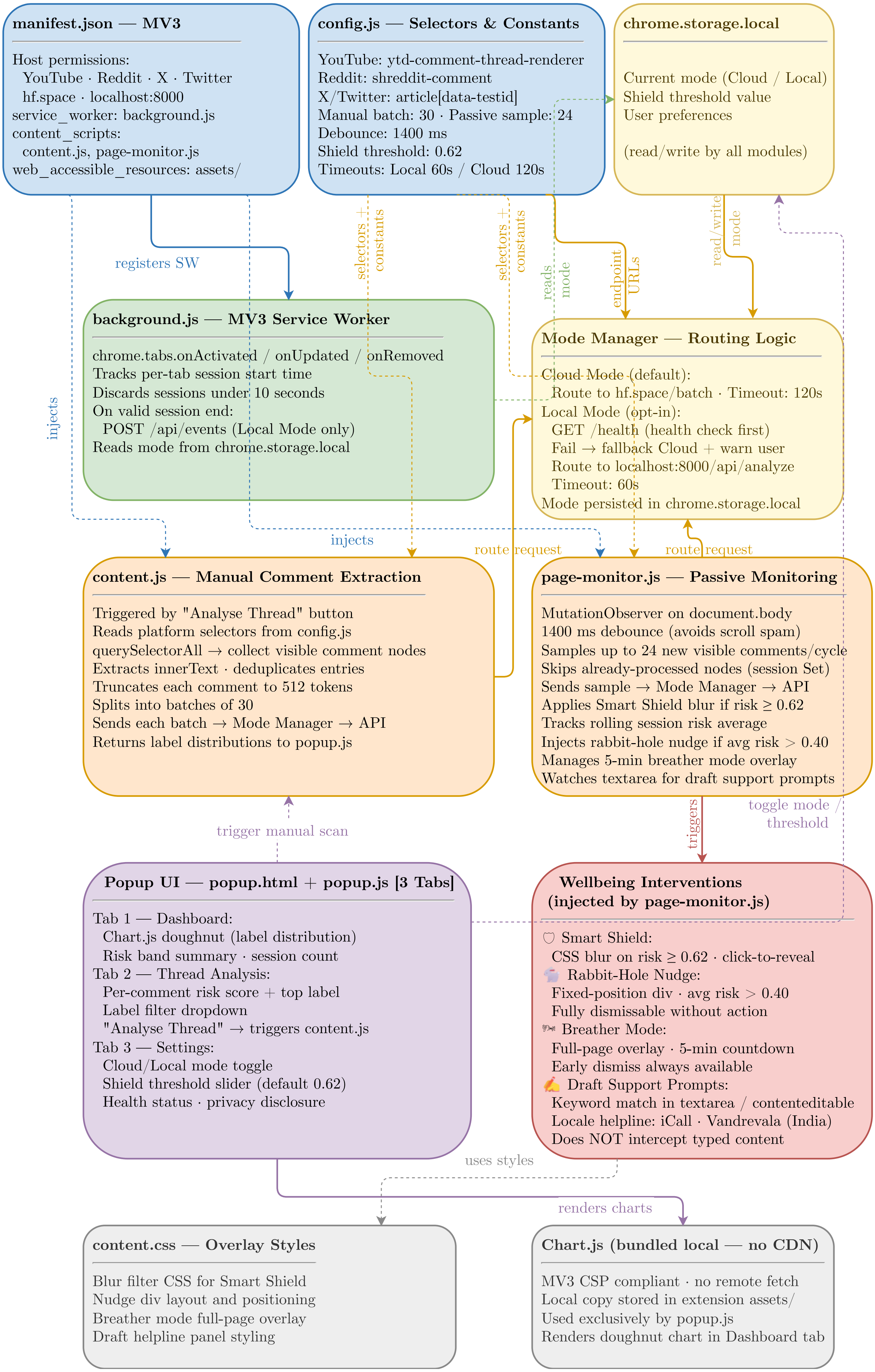

Figure 6.3: Extension Module Architecture

Band	Score Range	Description	Product Action
calm	0.00 - 0.39	Predominantly normal language	No intervention
guarded	0.40 - 0.67	Mixed or mildly elevated signal	Optional monitoring
high	0.68 - 0.83	Elevated distress-weighted signal	Shield if threshold met; dashboard flag
extreme	0.84 - 1.00	Strongly elevated multi-label signal	Shield applied; rabbit-hole nudge eligible

Table 6.4: Risk Score Thresholds and Bands

The backend also computes session volatility (flagged when max risk swing between consecutive events exceeds 0.45, or average swing exceeds 0.30) and an emotional-diet breakdown (label exposure percentages over the 7-day dashboard window).

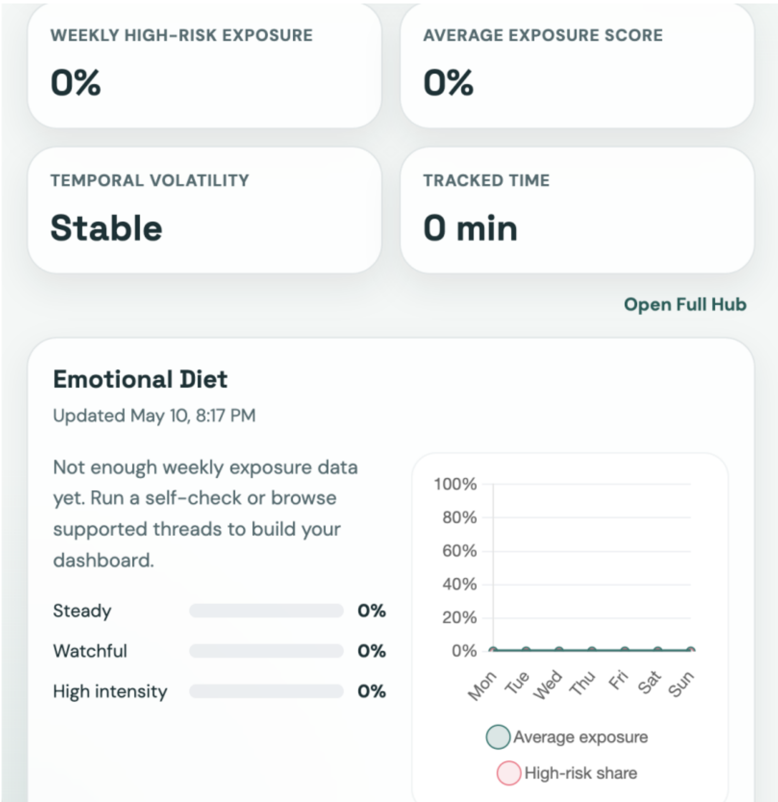

Figure 6.4: Emotional Diet Chart

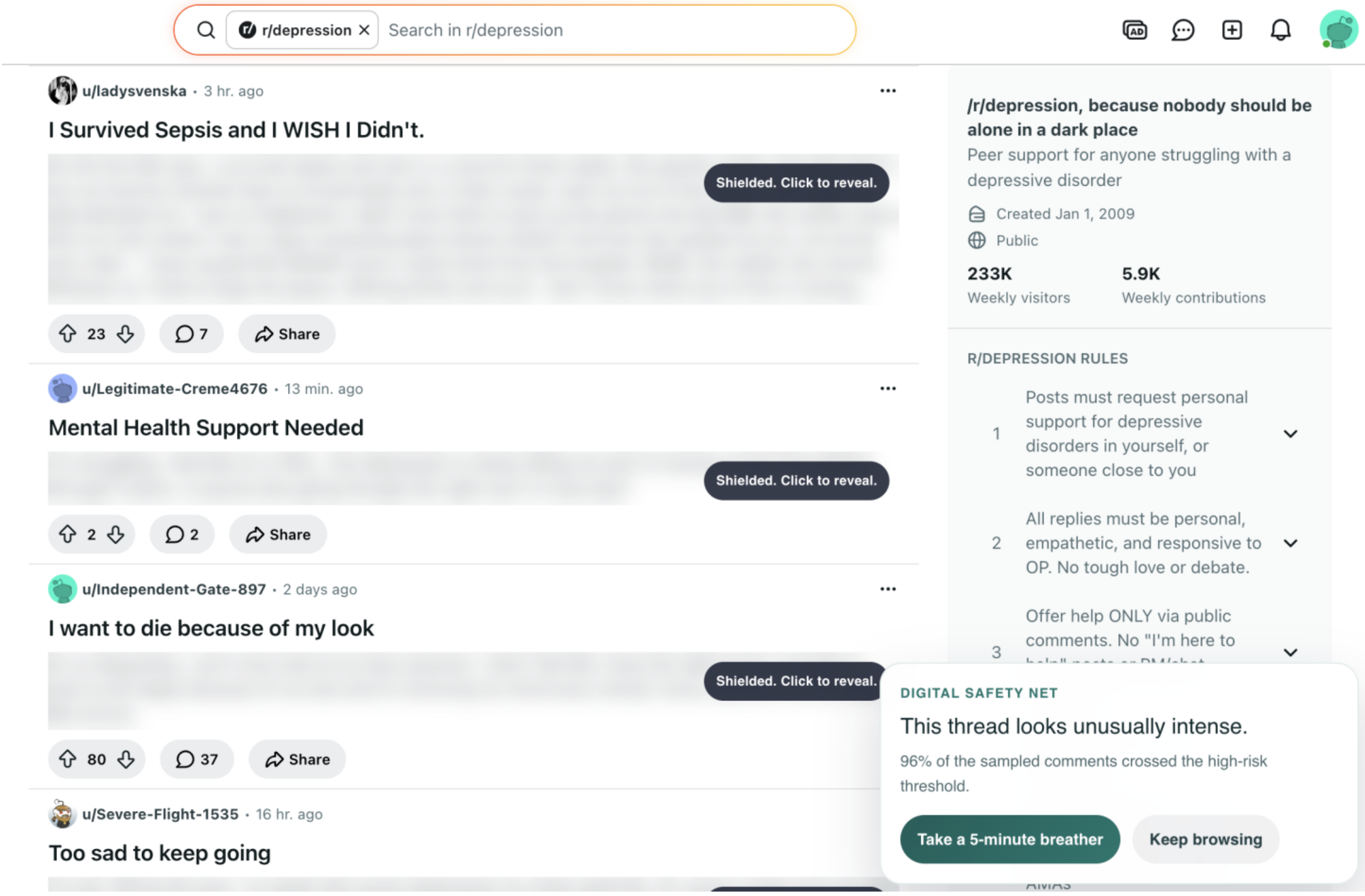

Figure 8.1: Smart Shield and Wellbeing Interventions Flow

8.2 Hugging Face Space API

The Space is hosted at https://ekam28-emotion-detector-api.hf.space as a Docker SDK Space (python:3.11-slim, Uvicorn on port 7860, CORS enabled). On startup, snapshot_download fetches the ONNX artifact from ekam28/emotion-detector-onnx pinned to a specific revision hash. ONNX Runtime InferenceSession is initialised with CPUExecutionProvider; AutoTokenizer is loaded from the same repository. /predict classifies a single text; /batch classifies up to 100. Both apply sigmoid over raw logits and return sorted label-probability pairs. Pydantic models enforce a 5,000 character maximum per text and a 100-item batch limit.